

Types of Polymorphism

Polymorphism can be divided into two types :

- ▶ **1) Static polymorphism** OR Compile time polymorphism OR Early binding
- ▶ **2) Dynamic Polymorphism** OR Runtime polymorphism OR Late binding

1) Static Polymorphism

The polymorphism which exist at the time of compilation is called Static OR compile time polymorphism.

In static polymorphism, compiler has very good idea that which method is invoked depending upon **METHOD PARAMETER**.

Here the binding of the method is done at compilation time so, it is known as early binding.

We can achieve static polymorphism by using Method Overloading concept.

Example of static polymorphism : Method Overloading.

2) Dynamic Polymorphism

The polymorphism which exist at runtime is called Dynamic polymorphism Or Runtime Polymorphism.

*Here compiler does not have any idea about method calling, at runtime JVM will decide which method will be invoked depending upon **CLASS TYPE**.

Here method binding is done at runtime so, it is also called Late Binding.

We can achieve dynamic polymorphism by using Method Overriding.

Method Overloading

Writing two or more methods in the same class or even in the super and sub class in such a way that the method name must be same but the argument must be different.

- ▶ While Overloading a method we can change the return type of the method.
- ▶ **If parameters are same but only method return type is different then it is not an overloaded method.**
- ▶ Method overloading is possible in the same class as well as super and sub class.
- ▶ While overloading the method the argument must be different otherwise there will be ambiguity problem.

Interview Questions & Notes

IQ : Can we overload the main method/static method ?

Yes, we can overload the main method OR static method but the execution of the program will start from main method which accept String [] array as a parameter.

Note :-

- ▶ The advantage of method overloading is same method name we can reuse for different functionality for refinement of the method.
- ▶ In `System.out.println()` or `System.out.print()`, `print()` and `println()` methods are best example for Method Overloading.

the 1990s, the number of people in the world who are under 15 years of age is expected to increase by 1.5 billion (United Nations 1994).

There is a growing awareness of the need to address the needs of children in the 1990s. The United Nations has developed the Convention on the Rights of the Child (1989) and the United Nations Children's Fund (UNICEF) has developed the Innocent Initiative (1990). The Innocent Initiative is a global campaign to improve the lives of children and to protect them from violence, abuse, and exploitation.

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